Solution Exercise 02 ROOO4E UPH/25

2. a. TF : $G(s) = \dfrac{1}{s^2+6s+58}$

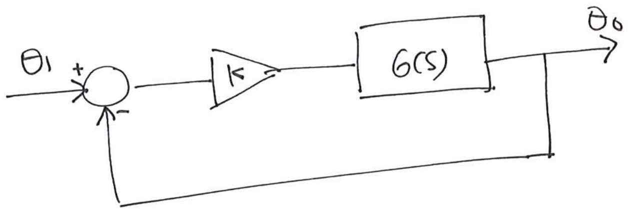
3. $\dot{x} = Ax + Bu = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u$

$y = Cx + Du = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} x$

$\dfrac{Y(s)}{U(s)} = \dfrac{3s+5}{s^2+4s+6} = \dfrac{3s+5}{(s+2)^2-(j\sqrt{2})^2}$

c. poles : $p_{1,2} = -2 \pm j\sqrt{2}$

System <u>asymptotically</u> stable

4. 

$G(s) = \dfrac{6}{s^2+4.2s+36}$

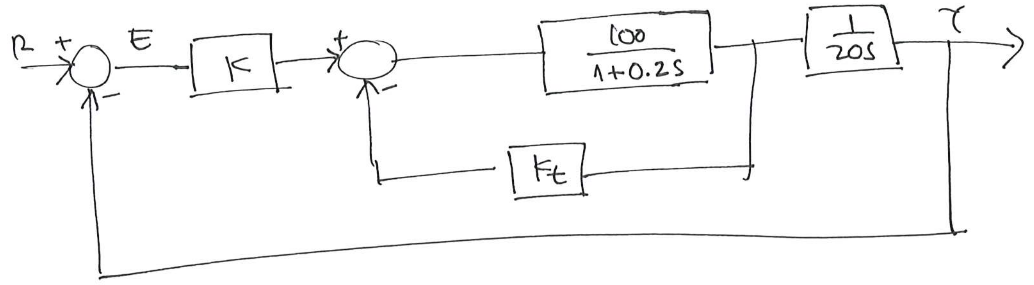
a. $T(s) = \dfrac{6}{s^2+4.2s+42}$

b. $e_{ssOL} = \dfrac{5}{6}$ w.r.t. unit step input

$e_{ssCL} = \dfrac{6}{7}$

c. Type of system: Type $\emptyset$, e_{ss} = constant (w.r.t. unit step input)

7.



a. $K = 1$, $K_t = 0$

System is asymptotically stable

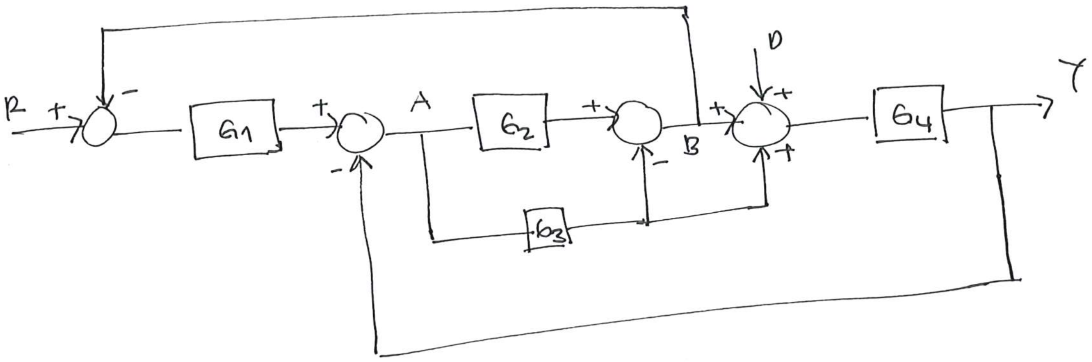
b. $R = \frac{1}{S^2}$ unit ramp

$$e_{ss} = \frac{5(1 + 100K_t)}{25K}$$

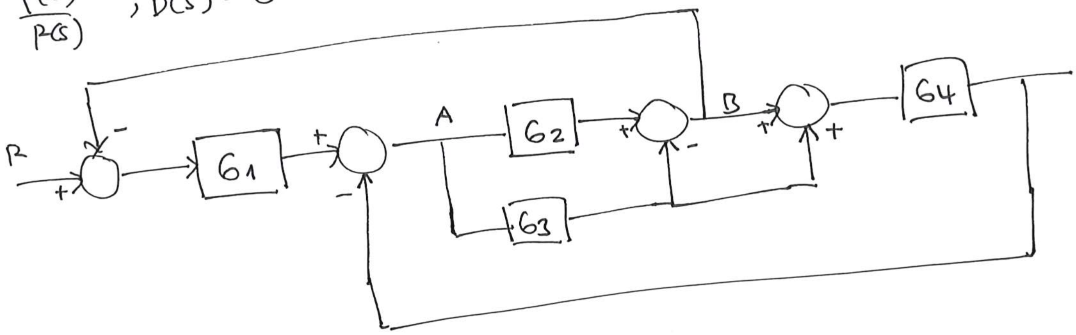
K_t >>> e_{ss} ↗

K >>> e_{ss} ↙

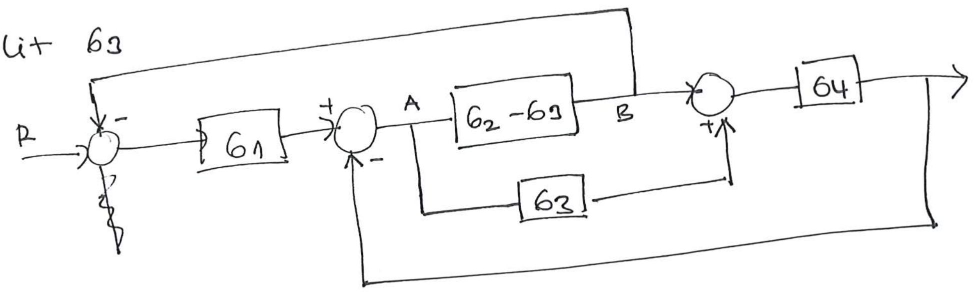
6.



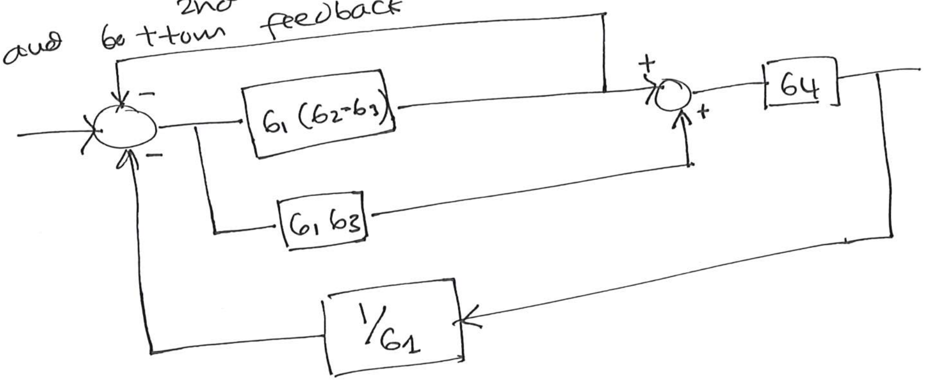
a. $\frac{Y(s)}{R(s)}$, $D(s) = 0$



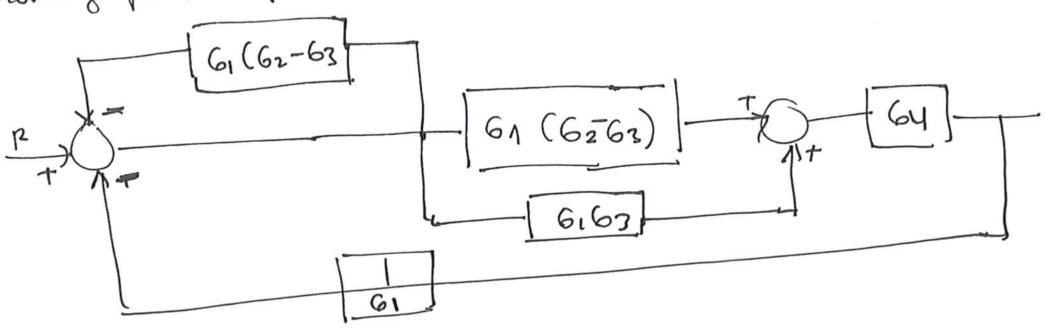
split G_3



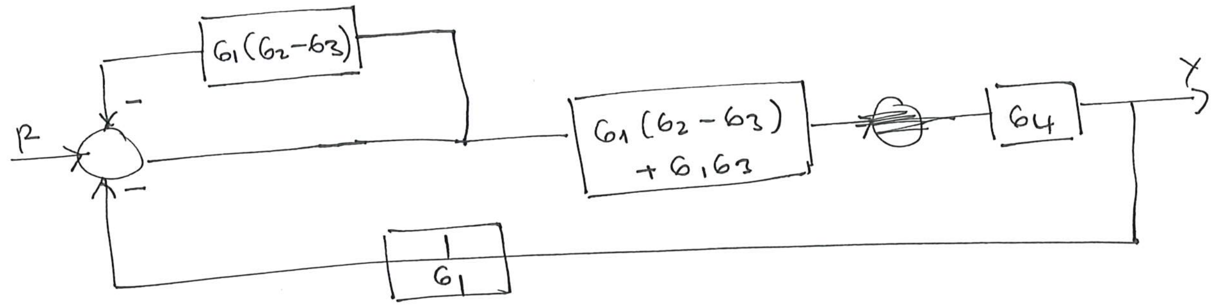
moving the 2nd summing point ~~behind~~ ahead a block, combining the top and bottom feedback



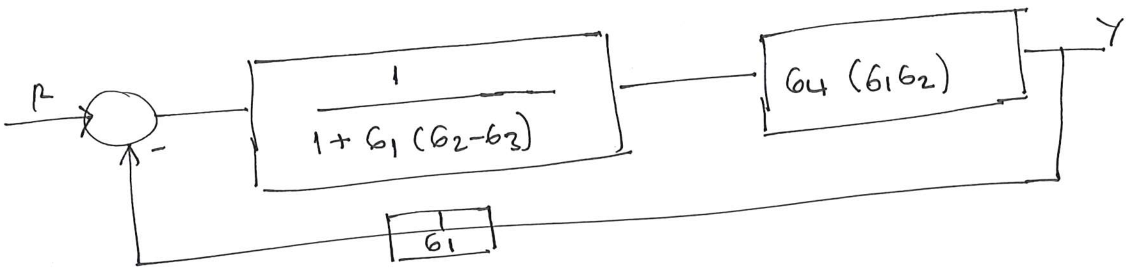
moving pick up point ahead of a block



Combine paraller blocks



Solve the top FB, and combine cascade blocks



~~Solve~~ Combine cascade blocks, then solve te FB

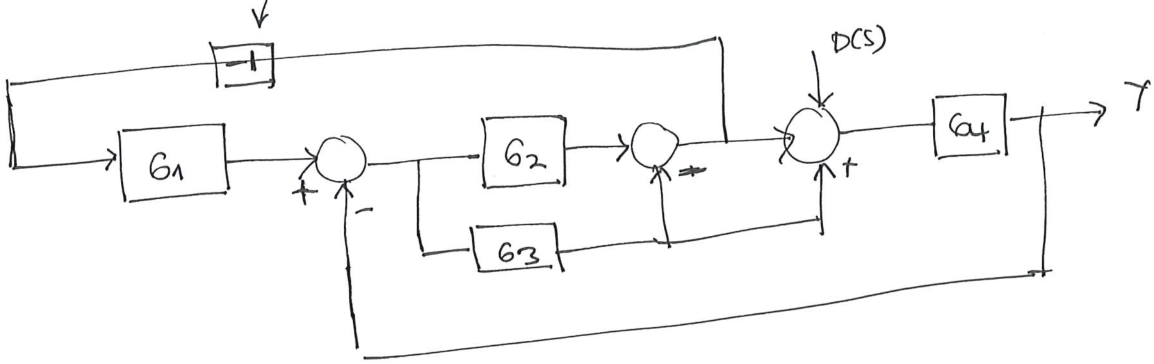
$$\frac{Y}{R} = \frac{G_4(G_1G_2)}{1+G_1(G_2-G_3) + G_4G_1G_2 \cdot \frac{1}{G_1}}$$

$$= \frac{G_1G_2G_4}{1+G_1(G_2-G_3)+G_2G_4}$$

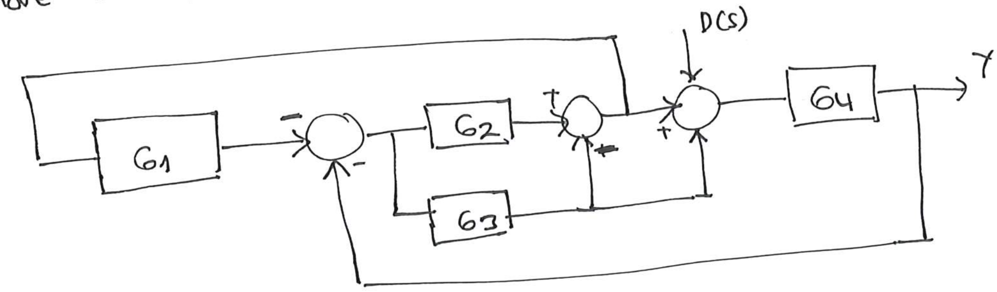
$$\frac{Y}{R} = \frac{G_1G_2G_4}{1+G_1G_2+G_2G_4-G_1G_3}$$

b. $\frac{Y(s)}{D(s)}$, $R(s) = 0$

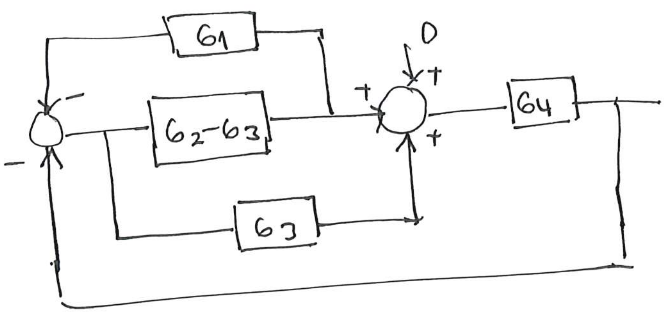
because of the ⊖ sign on the sum



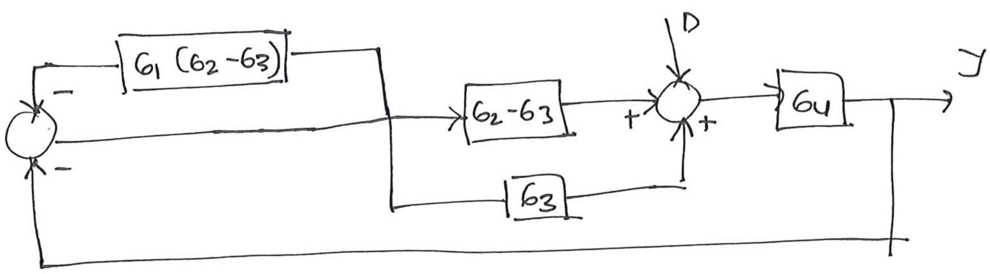
move the −1 to the sum ⊖



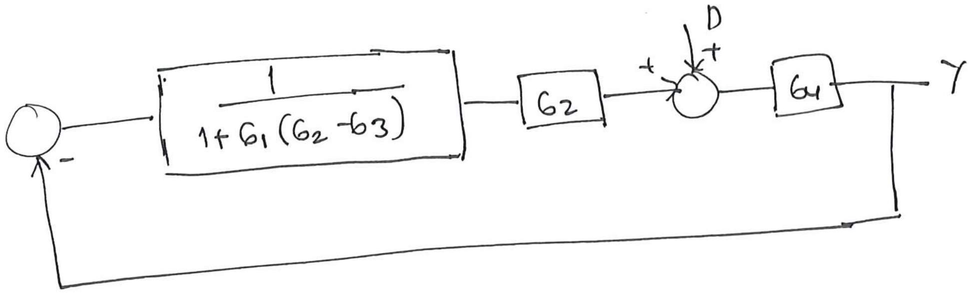
~~combine~~ split G3, move position of G1 to PO



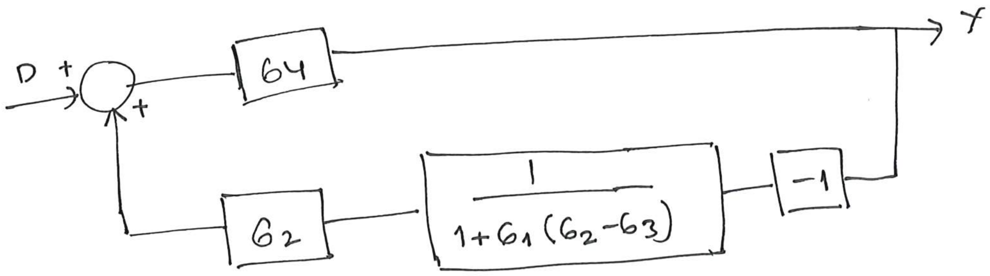
move ~~summing~~ pick off point ahead of a block



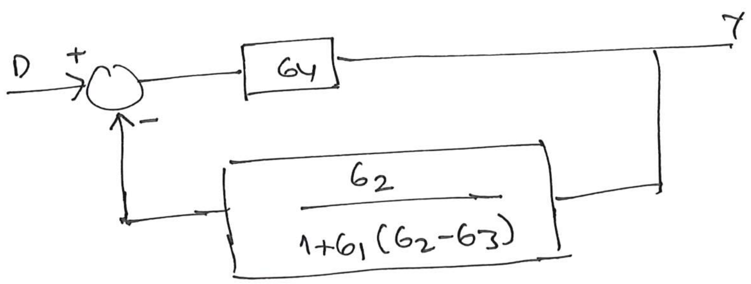
solve top FB, combine paraller blocks



rearrange the block



move -1 to the sum



solve FB

$$\frac{Y(s)}{D(s)} = \frac{G_4}{1 + \frac{G_2 G_4}{1+G_1(G_2-G_3)}}$$

$$= \frac{G_4(1+G_1(G_2-G_3))}{1+G_1(G_2-G_3)+G_2G_4}$$

$$\frac{Y(s)}{D(s)} = \frac{G_4 + G_1G_2G_4 - G_1G_3G_4}{1+G_1G_2+G_2G_4-G_1G_3}$$